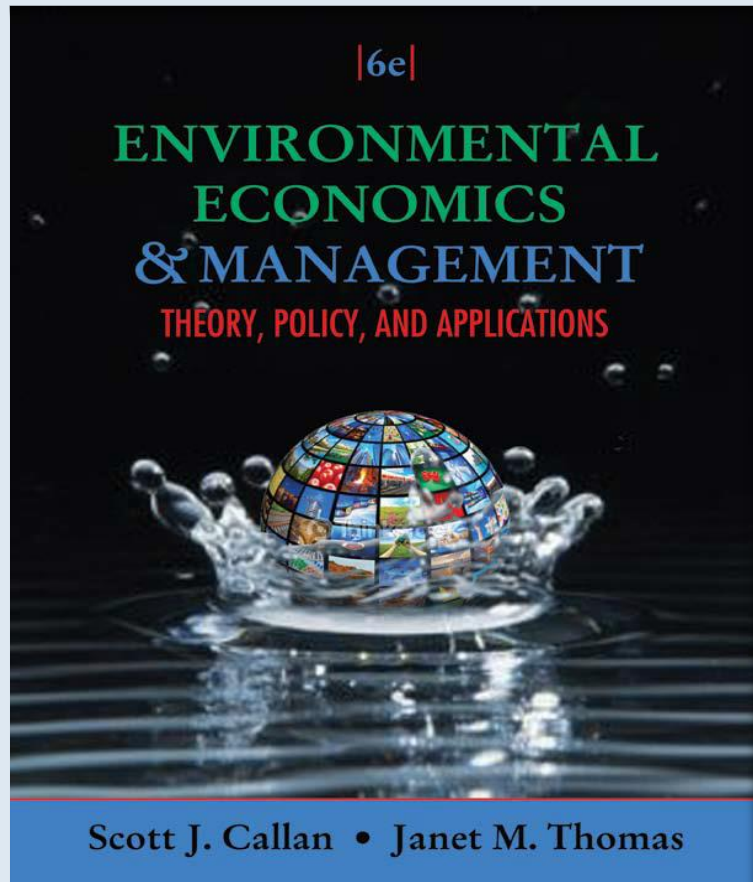




Lecture 13: Modeling the Market Failure

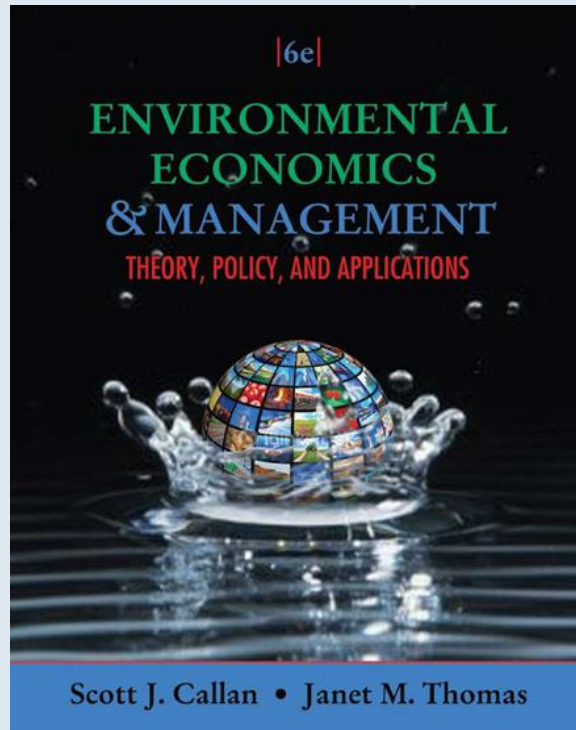
Environmental Economics and Management: Theory, Policy, and Applications 6e

by Scott J. Callan and Janet M. Thomas

Slides created by Janet M. Thomas

MODELING MARKET FAILURE

■ Chapter 3

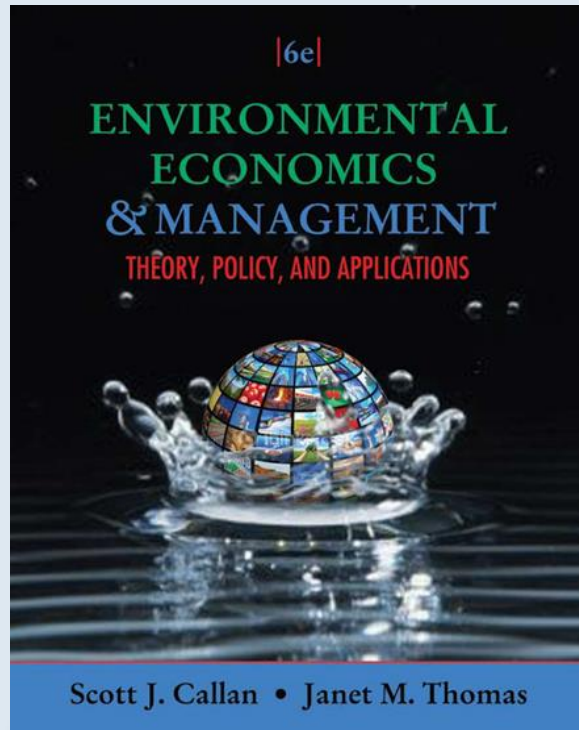


Environmental Pollution

A Market Failure

- **Market failure** is the result of an inefficient market condition
- Environmental problems are modeled as market failures using either the theory of public goods or the theory of externalities
 - If the market is defined as “environmental quality,” then the source of the market failure is that environmental quality is a **public good**
 - If the market is defined as the good whose production or consumption generates environmental damage, then the market failure is due to an **externality**

Public Goods Approach

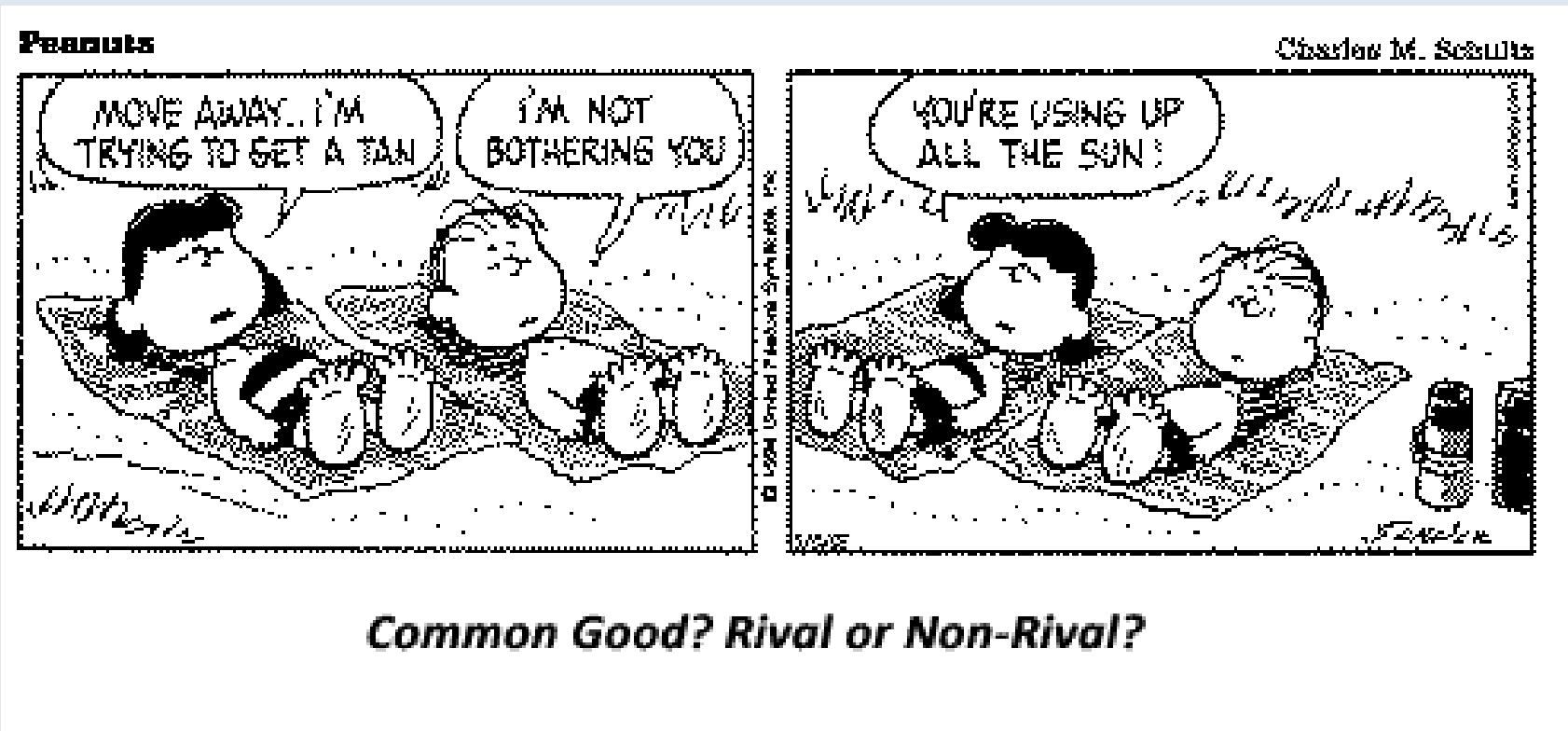


Environmental Quality

A Public Good

- A public good is a commodity that is **nonrival in consumption** and yields **nonexcludable** benefits
 - **Nonrivalness** – the characteristic of indivisible benefits of consumption such that one person's consumption does not preclude that of another
 - **Nonexcludability** – the characteristic that makes it impossible to prevent others from sharing in the benefits of consumption
- The relevant market definition is the public good – environmental quality, which possesses these characteristics

Example of Public Goods



Phot credit: <https://ednotesonline>

A Public Goods Market for Environmental Quality

- Public goods generate a market failure because the nonrivalness and nonexcludability characteristics prevent market incentives from achieving allocative efficiency
- Achieving allocative efficiency in a public goods market depends on the existence of well-defined supply and demand functions
 - But the public goods definition disallows the conventional derivation of market demand

Market Demand for a Public Good

- In theory, market D for a public good is found by *vertically summing* individual demands
 - Vertical sum because we must ask consumers “*What **price** would you be willing to pay for each quantity of the public good?*”
- But consumers are unwilling to reveal their WTP because they can share in consuming the public good even when purchased by someone else
 - Due to the nonrival and nonexcludability characteristics
- This problem is called **nonrevelation of preferences**, which arises due to **free-ridership**

Market Demand for a Public Good

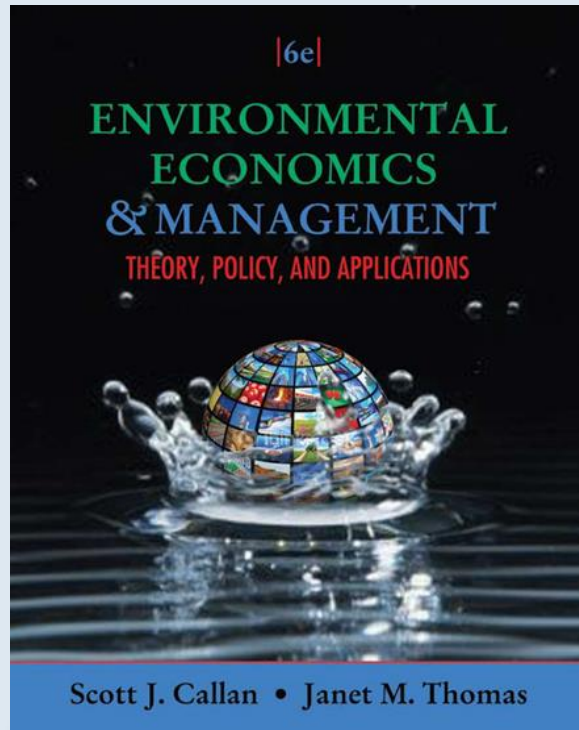
- Result is that market demand is undefined
- In addition, lack of awareness of environmental problems (i.e., imperfect information) exacerbates the problem
- Consequently, allocative efficiency cannot be achieved without third-party intervention

Solution to Public Goods Dilemma

Government Intervention

- Government might respond through direct provision of public goods
- Government might use political procedures and voting rules to identifying society's preferences about public goods

Externality Approach



Environmental Problems

A Negative Externality

- An **externality** is a spillover effect associated with production or consumption that extends to a third party outside the market
 - **Negative externality** – an external effect that generates costs to a third party
 - **Positive externality** – an external effect that generates benefits to a third party

Environmental Problems

A Negative Externality

- Environmental economists are interested in externalities that damage the atmosphere, water supply, natural resources, and overall quality of life
- To model these **environmental externalities**, the relevant market must be defined as the good whose production or consumption generates environmental damage outside the market transaction

Relationship Between Public Goods and Externalities

- Although public goods and externalities are not the same concept, they are closely related
 - If the externality affects a broad segment of society and if its effects are nonrival and nonexcludable, the externality is itself a public good
 - If the externality affects a narrower group of individuals or firms, those effects are more properly modeled as an externality

Modeling a Negative Environmental Externality

- Define the market as refined petroleum
 - Assume the market is competitive
 - Supply is the marginal *private* cost (MPC)
 - Demand is the marginal *private* benefit (MPB)
 - Production generates pollution, modeled as a marginal *external* cost (MEC)
- **Problem:** Producers (refineries) have no incentive to consider the externality
- **Result:** Competitive solution is inefficient

Finding a Competitive Solution

Refined Petroleum Market (text example)

- S: $P = 10.0 + 0.075Q$
- D: $P = 42.0 - 0.125Q$, where
Q is thousands of barrels per day
- Since S is MPC and D is MPB, rewrite as:
 $MPC = 10.0 + 0.075Q$
 $MPB = 42.0 - 0.125Q$
- Find the competitive solution and analyze

Competitive Solution

- Set $MPB = MPC$

$$42.0 - 0.125Q = 10.0 + 0.075Q$$

- Solve:

$$Q_C = 160 \text{ thousand} \quad P_C = \$22 \text{ per barrel}$$

- Analysis:

- This ignores external costs from contamination
- Allocative efficiency requires P to equal *all* MC
- MPC undervalues opportunity costs of production; Q_C is too high; P_C is too low

Finding an Efficient Solution

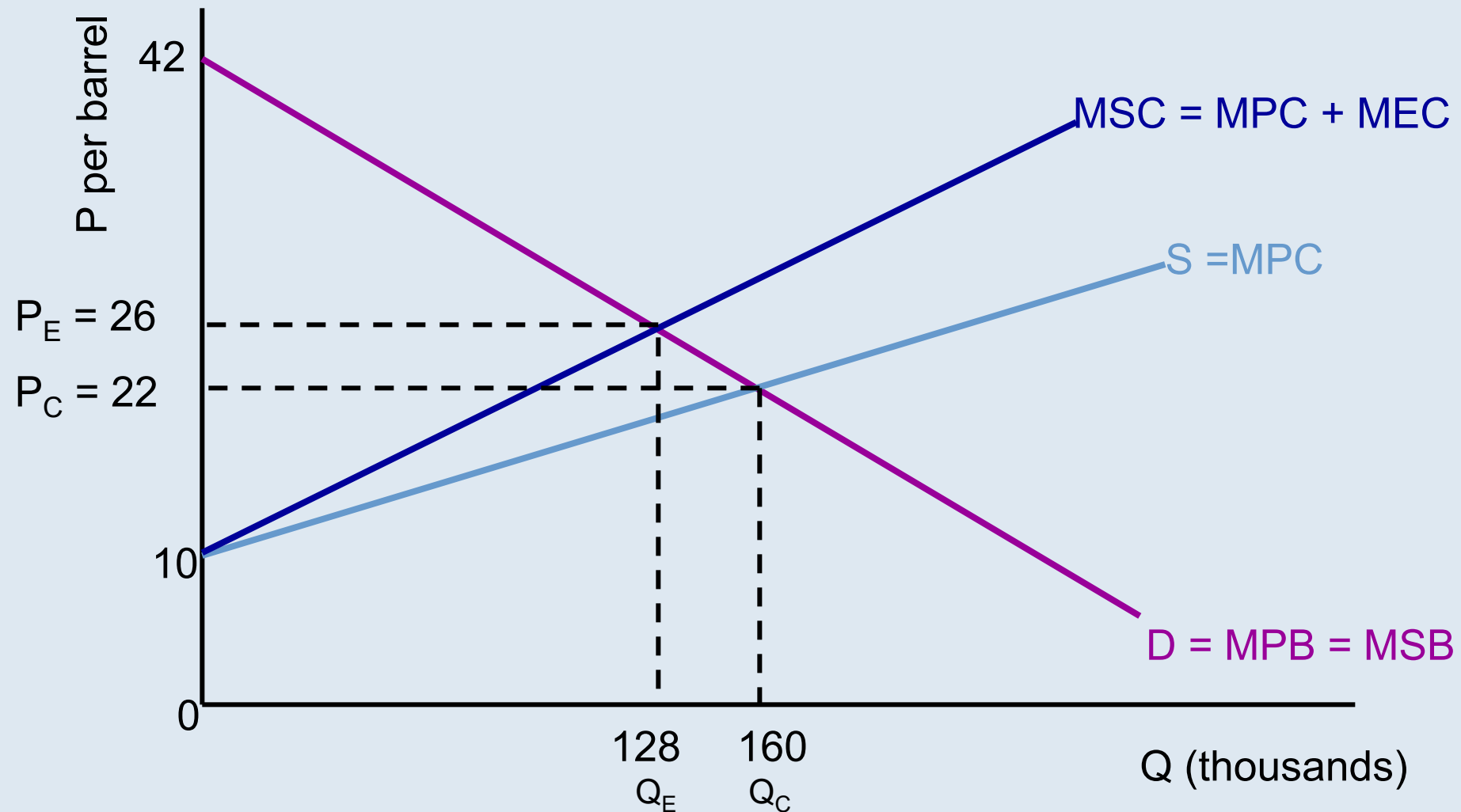
Refined Petroleum Market

- Let Marginal External Cost (MEC) = $0.05Q$
- Marginal Social Cost (MSC) = $MPC + MEC$
 - $MSC = 10.0 + 0.075Q + 0.05Q$
 $= 10.0 + 0.125Q$
- Marginal Social Benefit (MSB) = $MPB + MEB$
 - Assuming no external benefits, $MEB = 0$, so $MSB = MPB$
- Find the efficient solution; show graphically

Efficient Solution

- Set $MSC = MSB$
 - $10.0 + 0.125Q = 42.0 - 0.125Q$
 - Solving: $Q_E = 128$ thousand $P_E = \$26/\text{barrel}$
- Observe: In the presence of an externality, market forces *cannot* determine an efficient outcome

MSC, MPC, MPB Graph



Observations

- Results of negative externality
 - Q_C is too high, i.e., overallocation of resources
 - P_C is too low, since MEC is not captured by market transaction

Comparing the Equilibria

- An alternative way to analyze the two equilibria is to examine the corresponding levels of $M\pi$.

Competitive equilibrium: $MPB = MPC$
 $MPB - MPC = 0$
 $M\pi = 0$

Efficient equilibrium: $MSB = MSC$
 $MPB + MEB = MPC + MEC$
 $MPB - MPC = MEC \quad (\text{since } MEB = 0)$
 $M\pi = MEC$

Comparing the Equilibria

Using $M\pi$ and MEC

- Competitive firm maximizes π where
 - $MPB = MPC$, or where $MPB - MPC = 0$, or
 - $M\pi = 0$
 - ▣ since $MPB - MPC = M\pi$ by definition
- Efficient firm produces where
 - $MSB = MSC$ or $MPB + MEB = MPC + MEC$
 - or $MPB - MPC = MEC$, if $MEB = 0$, so...
 - $M\pi = MEC$

Model

Refined Petroleum Market

- $M\pi = MPB - MPC$
 $= (42 - 0.125Q) - (10 + 0.075Q)$ so
- $M\pi = 32 - 0.2Q$
- $MEC = 0.05Q$
- Find the competitive and efficient equilibria using these equations

Solution

■ Competitive solution

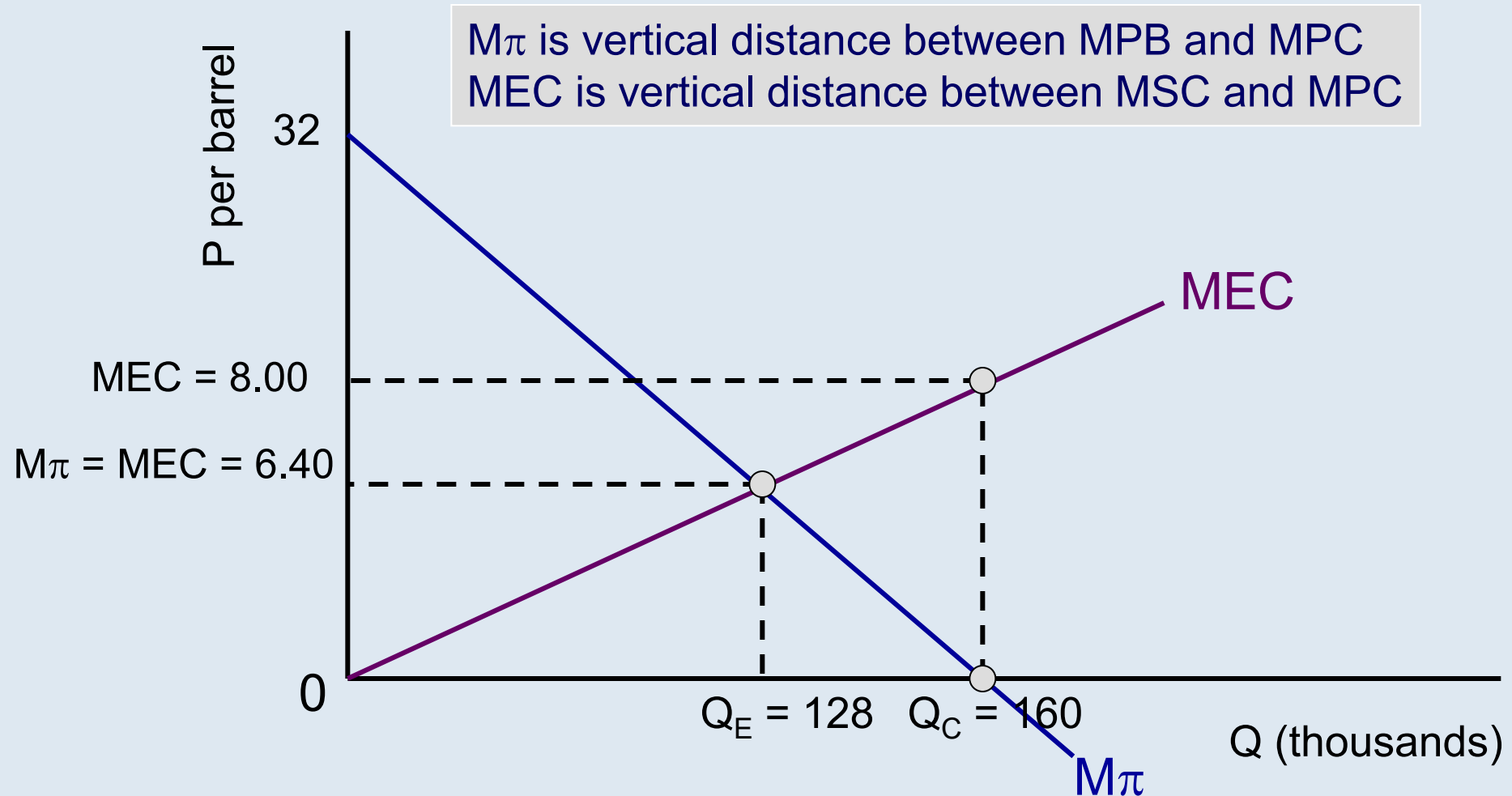
- Set $M\pi = 0$, or $32 - 0.2Q = 0$, so $Q_C = 160$
- Find P by substituting into MPB or MPC
 - ▢ Using MPB, $P_C = 42 - 0.125(160) = 22$

■ Efficient solution

- Set $M\pi = MEC$, or $32 - 0.2Q = 0.05Q$, so $Q_E = 128$
- Find P by substituting into MPB or MPC
 - ▢ Using MPB, $P_E = 42 - 0.125(128) = 26$

$M\pi$, MEC Graph

Refined Petroleum Market



Analysis

- $Q_C = 160$ thousand
 - At this point, $MEC = \$8.00$ per barrel
 - Note $M\pi \neq MEC \Rightarrow$ not efficient
- $Q_E = 128$ thousand
 - At this point, $MEC = M\pi = \$6.40$ per barrel
- Efficiency would improve if output were restricted by 32 thousand (i.e., $160 - 128$)

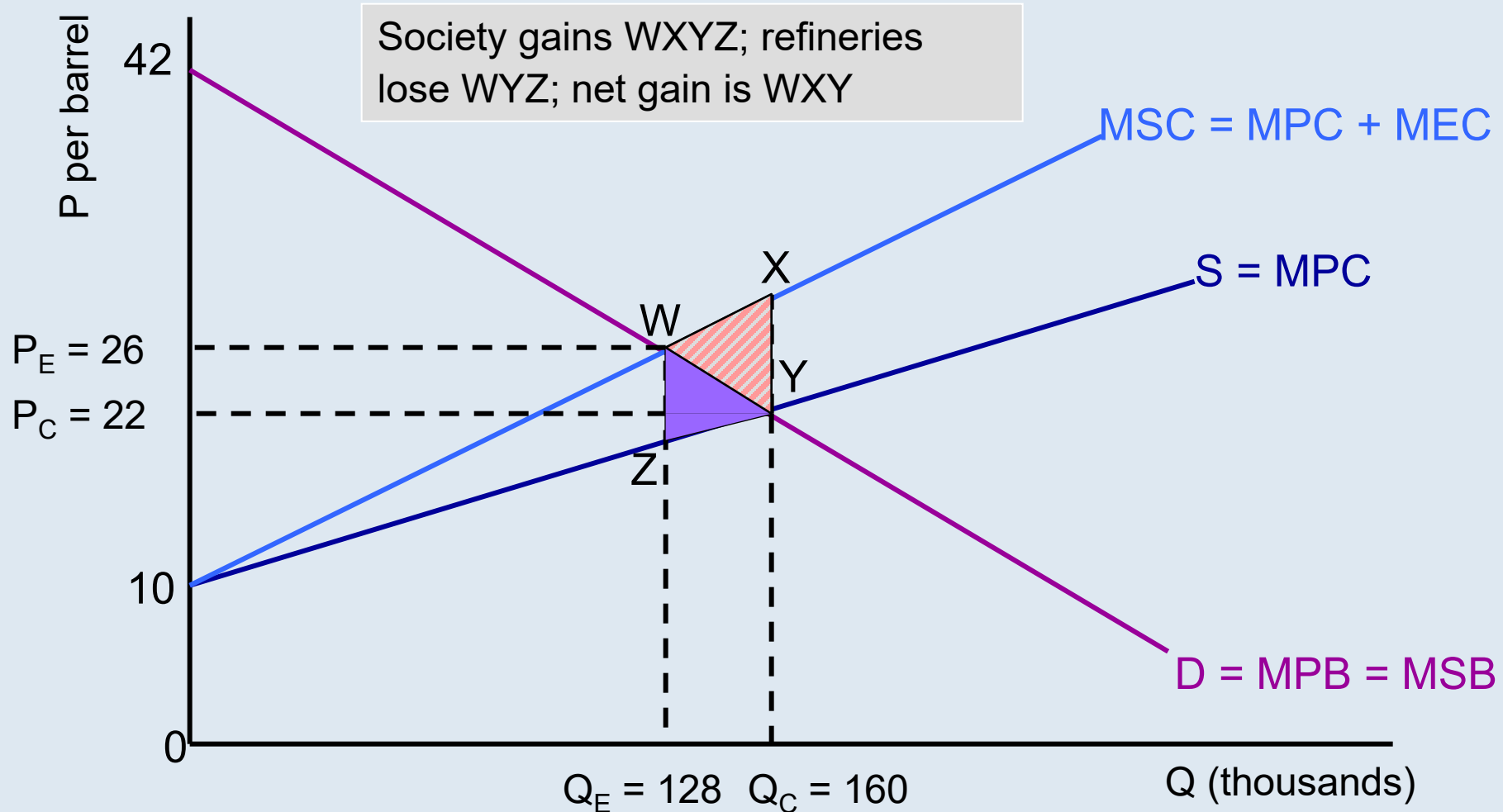
Measuring Society's Net Gain

From Restoring Efficiency

- As Q falls from 160 to 128:
 - Refineries lose π measured as $M\pi$ (or excess of MPB over MPC) for each unit of Q contracted
 - ▣ Defines area **WYZ**
 - Society gains accumulated reduction in MEC for each unit of Q contracted
 - ▣ Defines area **WXYZ**
 - Net gain = Area WXYZ - Area WYZ = Area WXY

Measuring Society's Net Gain

Refined Petroleum Market



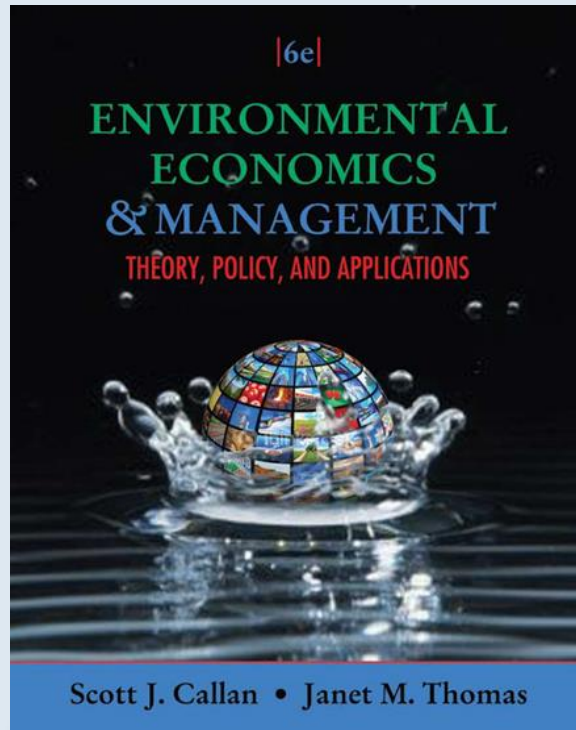
Important Observations

- Both externality and public goods models show inefficiency of private market solution, i.e., market failure
- Underlying source of failure is **absence of property rights**
 - Recall Boston Harbor application



The screenshot displays the MWRA online website. On the left is a vertical navigation menu with links: Home, Water System, Sewer System, Harbor and Bay (highlighted), School Program, About MWRA, Doing Business with MWRA, and Contact MWRA. The main content area is titled "Boston Harbor and Massachusetts Bay MWRA Environmental Quality Department". Below this title are four tabs: "What's New", "Regulatory Reporting", "Technical Reports", and "Archive". The "What's New" tab is active, showing a section titled "Boston Harbor". The text in this section describes the 50 square miles of the Boston Harbor estuary, mentioning tributary rivers and eight main watersheds. It highlights the "Boston Harbor Project" which improved sewage treatment and moved wastewater treatment plant discharges from Boston Harbor to Massachusetts Bay. It also mentions that the MWRA collects water quality data at more than 70 locations and that the results show how Boston Harbor has become cleaner since effluent and solids discharges were removed. A photograph of a sailboat in the harbor with the city skyline in the background is shown on the right side of the text.

Absence of Property Rights



■ The Coase Theorem

Ronald Coase, Nobel Laureate, 1991

Property Rights

- Valid claims to a good or resource that permit the use and transfer of ownership through sale
- For environmental goods, it's unclear who “owns” rights
- Economics says it's the *absence* of rights that matters, not who possesses them

Coase Theorem

- Proper assignment of property rights to any good, even if externalities are present, will allow bargaining between parties such that efficient solution results, regardless of who holds rights
 - Assumes costless transactions
 - Assumes damages are accessible and measurable

Building the Model

Refined Petroleum Market

- To illustrate the Coase Theorem, let's revisit our market model of refined petroleum to see how the assignment of property rights and bargaining can restore efficiency.
- Refineries use the river to release chemicals as an unintended by-product of production
 - Objective: to maximize π
- Recreational users use the river for swimming and boating
 - Objective: to maximize utility

Bargaining When Rights Belong to Refineries

- Recreational users are willing to pay (WTP) refineries for each unit of Q not produced
 - Will pay *up to* the negative effect on utility (MEC)
- Refineries are willing to accept payment not to produce
 - Will accept payment *greater than* their loss in profit from contracting production ($M\pi$)

Recreational users: Willing to offer a payment (ρ) such that $\rho < (MSC - MPC)$

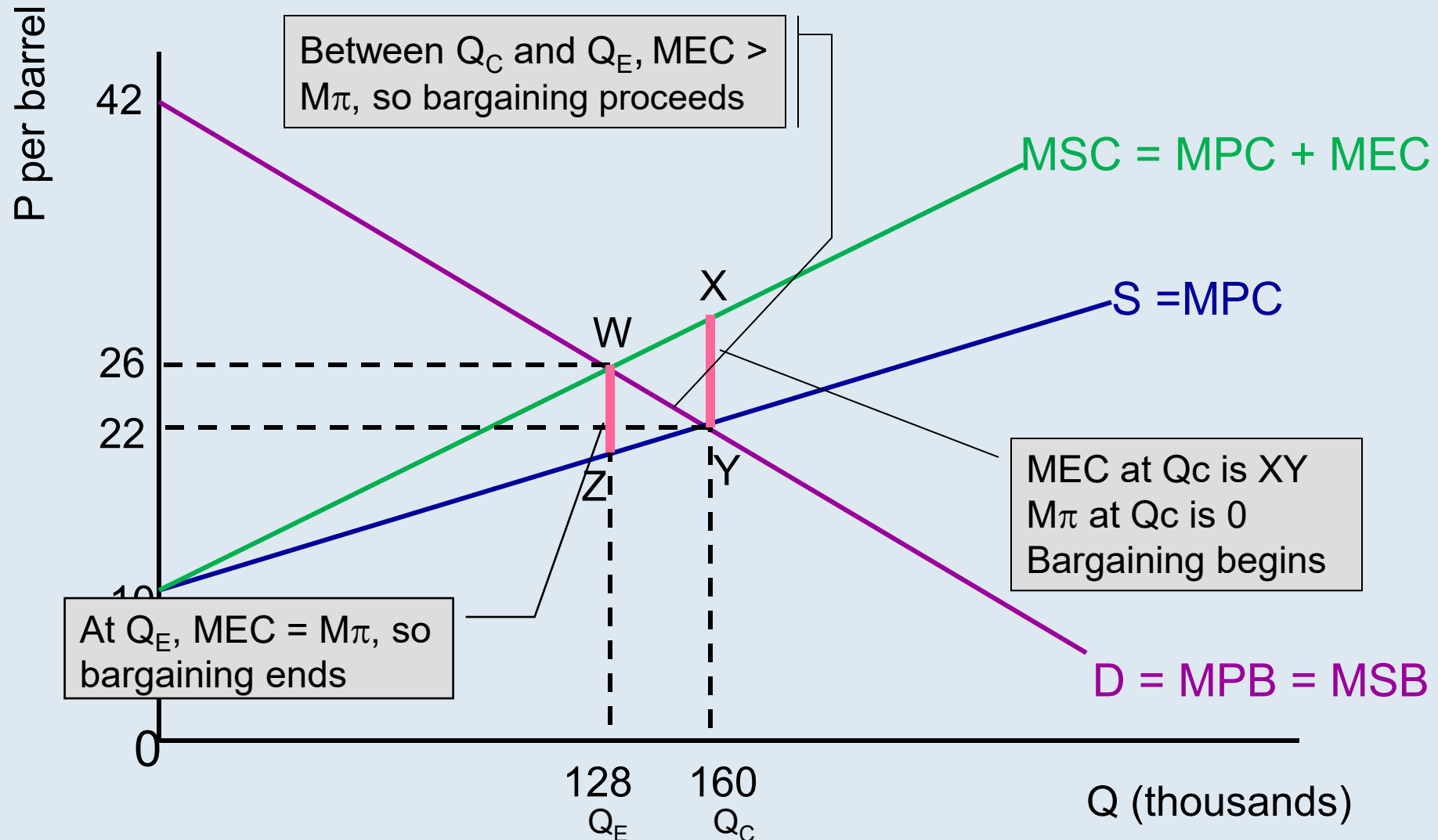
Refineries: Willing to accept a payment (ρ) such that $\rho > (MPB - MPC)$

Bargaining When Rights Belong to Refineries

- Initial point is Q_c , since the refineries, who own the rights, would choose this point
- Recreational users:
 - Willing to offer a payment ρ
 - $\rho < (MSC - MPC)$, or $\rho < MEC$
- Refineries:
 - Willing to accept payment ρ
 - $\rho > (MPB - MPC)$, or $\rho > M\pi$

$$\begin{array}{ccc} (MSC - MPC) & > \rho > & (MPB - MPC), \text{ or equivalently,} \\ MEC & > \rho > & M\pi \end{array}$$

Bargaining Process



Bargaining Process

- Bargaining should continue as long as:
 $(MSC - MPC) > \rho > (MPB - MPC)$ or
 $MEC > \rho > M\pi$
- At Q_C : Refineries' $M\pi = 0$, but $MEC > 0$, (distance XY)
 - Since $MEC > M\pi$, bargaining begins
- Between Q_C and Q_E , same condition holds
- At Q_E : $MEC = M\pi$, (distance WZ); output reductions beyond this point are infeasible, since $M\pi > MEC$

Bargaining When Rights Belong to Recreational Users

- Bargaining will proceed analogously
- The starting point at which bargaining begins, however, is different. If the recreational users hold the rights to the river, then technically, the refineries cannot produce at all unless they pay for the rights to pollute. Thus, we assume that the point at which bargaining begins is where $Q=0$

Refineries:	Willing to offer a payment ρ such that $\rho < (MPB - MPC)$
Recreational users:	Willing to accept a payment ρ such that $\rho > (MSC - MPC)$

Bargaining When Rights Belong to Recreational Users

- Therefore, there is opportunity for bargaining to proceed as long as the following condition holds:

$$\begin{array}{ccc} (MPB - MPC) & > \rho > & (MSC - MPC), \text{ or equivalently,} \\ M\pi & > \rho > & MEC \end{array}$$

- An efficient outcome can be realized without government intervention